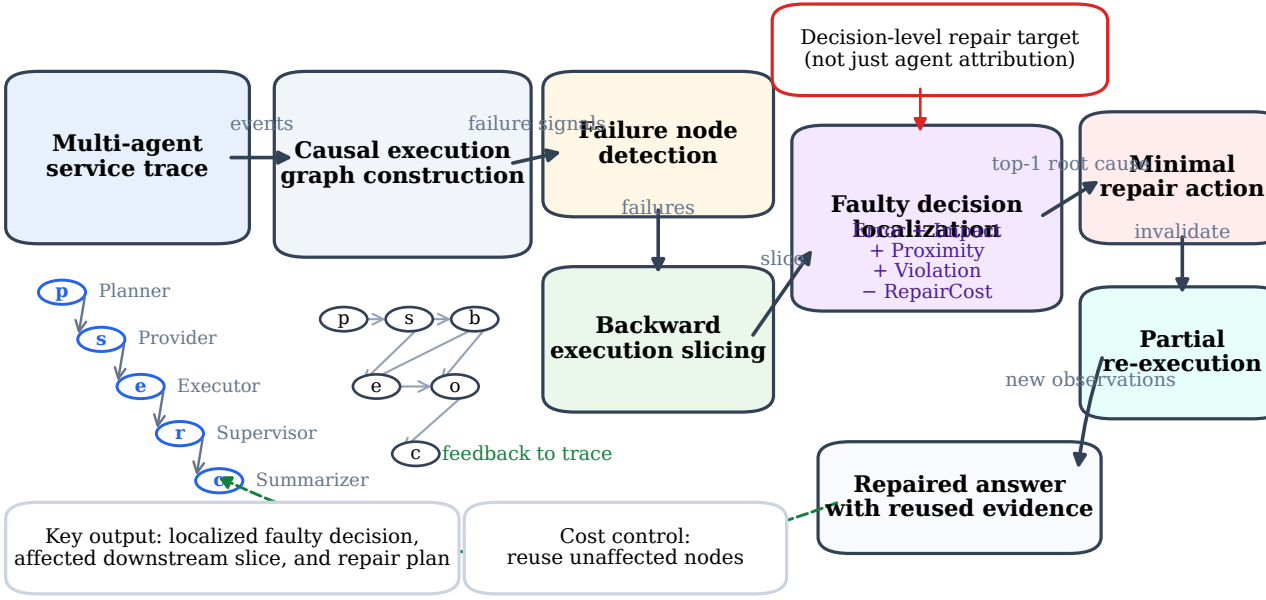
Multi-agent service trace
events
Causal execution graph construction
failure signals
Failure node detection
failures
Backward execution slicing
slice
Decision-level repair target (not just agent attribution)
Faulty decision localization
+ Proximity
+ Violation
− RepairCost
top-1 root cause
Minimal repair action
invalidate
Partial re-execution
new observations
Repaired answer with reused evidence
p Planner
s Provider
e Executor
r Supervisor
c Summarizer
p
s
b
e
o
c
feedback to trace
Key output: localized faulty decision, affected downstream slice, and repair plan
Cost control: reuse unaffected nodes